

apple1

April 7, 2026

```
[94]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import statsmodels.api as sm
from statsmodels.sandbox.regression.gmm import IV2SLS
import statsmodels.formula.api as smf
```

```
[95]: apple_panel = pd.read_csv('apple_panel_washington.csv')
apple_panel = apple_panel[(apple_panel['year'] >= 2010) & (apple_panel['year'] ≤
    ↪ ≤ 2024)].copy()
apple_panel['trend'] = apple_panel['year'] - 2009

print(apple_panel.head(10))
```

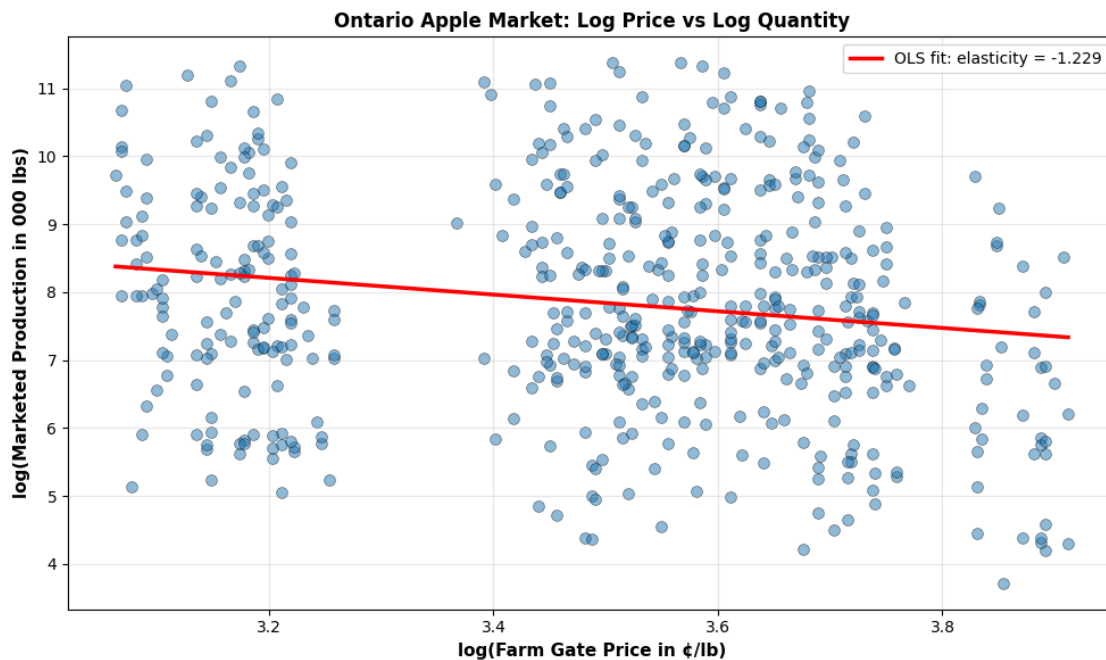
	county	year	log_prod	log_price	log_wa_price	trend
0	Brant	2024.0	8.513988	3.701302	3.617882	15.0
1	Chatham-Kent	2024.0	8.735364	3.608212	3.617882	15.0
2	Elgin	2024.0	10.243881	3.681351	3.617882	15.0
3	Essex	2024.0	10.130623	3.589059	3.617882	15.0
4	Haldimand-Norfolk	2024.0	10.788122	3.678829	3.617882	15.0
5	Hamilton	2024.0	8.324579	3.686376	3.617882	15.0
6	Lambton	2024.0	9.304832	3.589059	3.617882	15.0
7	Middlesex	2024.0	9.504129	3.637586	3.617882	15.0
8	Niagara	2024.0	9.649498	3.645450	3.617882	15.0
9	Oxford	2024.0	7.757479	3.650398	3.617882	15.0

```
[96]: import matplotlib.pyplot as plt
import numpy as np

plt.figure(figsize=(10, 6))
plt.scatter(apple_panel['log_price'], apple_panel['log_prod'], alpha=0.5, s=50,
    ↪ edgecolors='black', linewidth=0.5)

x_range = np.linspace(apple_panel['log_price'].min(), apple_panel['log_price'].
    ↪ max(), 100)
y_fitted = 12.1405 + (-1.2286) * x_range
plt.plot(x_range, y_fitted, color='red', linewidth=2.5, label=f'OLS fit:
    ↪ elasticity = -1.229')
```

```
plt.xlabel('log(Farm Gate Price in €/lb)', fontsize=11, fontweight='bold')
plt.ylabel('log(Marketed Production in 000 lbs)', fontsize=11,
↳fontweight='bold')
plt.title('Ontario Apple Market: Log Price vs Log Quantity', fontsize=12,
↳fontweight='bold')
plt.legend(fontsize=10)
plt.grid(True, alpha=0.3)
plt.tight_layout()
plt.show()
```



1 Specification 1

Naive OLS.

```
[97]: X1 = sm.add_constant(apple_panel[['log_price']])
y = apple_panel['log_prod']
model1 = sm.OLS(y, X1).fit()
print(model1.summary())
```

OLS Regression Results

Dep. Variable:	log_prod	R-squared:	0.030
Model:	OLS	Adj. R-squared:	0.028
Method:	Least Squares	F-statistic:	16.53

Date: Tue, 07 Apr 2026 Prob (F-statistic): 5.51e-05
Time: 20:00:30 Log-Likelihood: -1025.8
No. Observations: 543 AIC: 2056.
Df Residuals: 541 BIC: 2064.
Df Model: 1
Covariance Type: nonrobust

	coef	std err	t	P> t	[0.025	0.975]
const	12.1405	1.058	11.471	0.000	10.061	14.220
log_price	-1.2286	0.302	-4.065	0.000	-1.822	-0.635
Omnibus:	15.640	Durbin-Watson:	1.587			
Prob(Omnibus):	0.000	Jarque-Bera (JB):	8.188			
Skew:	0.064	Prob(JB):	0.0167			
Kurtosis:	2.412	Cond. No.	58.2			

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

2 Specification 2

With County FE.

```
[98]: county_dummies = pd.get_dummies(apple_panel['county'], drop_first=True).
      ↪ astype(int)
X2 = sm.add_constant(pd.concat([apple_panel[['log_price']], county_dummies],
      ↪ axis=1))
model2 = sm.OLS(y, X2).fit()
print(model2.summary())
```

OLS Regression Results

```
=====
Dep. Variable:          log_prod    R-squared:          0.928
Model:                  OLS         Adj. R-squared:       0.923
Method:                 Least Squares    F-statistic:       170.8
Date:                  Tue, 07 Apr 2026    Prob (F-statistic): 1.80e-261
Time:                  20:00:30          Log-Likelihood:    -319.85
No. Observations:      543              AIC:              717.7
Df Residuals:          504              BIC:              885.3
Df Model:               38
Covariance Type:       nonrobust
=====
```

	coef	std err	t	P> t
[0.025	0.975]			

const		8.9758	0.358	25.062	0.000
8.272	9.679				
log_price		-0.9665	0.086	-11.260	0.000
-1.135	-0.798				
Brant		2.6138	0.234	11.179	0.000
2.154	3.073				
Bruce		1.7388	0.234	7.438	0.000
1.280	2.198				
Chatham-Kent		2.9191	0.234	12.487	0.000
2.460	3.378				
Dufferin		0.0656	0.234	0.281	0.779
-0.394	0.525				
Durham		4.5894	0.234	19.628	0.000
4.130	5.049				
Elgin		4.4752	0.234	19.143	0.000
4.016	4.934				
Essex		4.4013	0.234	18.828	0.000
3.942	4.861				
Frontenac		-0.0959	0.236	-0.406	0.685
-0.560	0.368				
Grey		5.4221	0.234	23.196	0.000
4.963	5.881				
Haldimand-Norfolk		5.0102	0.234	21.431	0.000
4.551	5.469				
Halton		2.2458	0.234	9.606	0.000
1.787	2.705				
Hamilton		3.1050	0.234	13.280	0.000
2.646	3.564				
Hastings		1.6184	0.234	6.922	0.000
1.159	2.078				
Huron		1.4499	0.234	6.202	0.000
0.991	1.909				
Kawartha Lakes		-0.1152	0.248	-0.465	0.642
-0.602	0.372				
Lambton		3.5872	0.234	15.346	0.000
3.128	4.046				
Lanark		0.3401	0.234	1.455	0.146
-0.119	0.800				
Leeds & Grenville		1.8669	0.234	7.983	0.000
1.407	2.326				
Lennox & Addington		1.4123	0.236	5.985	0.000
0.949	1.876				
Manitoulin		0.1359	0.234	0.581	0.561
-0.324	0.595				
Middlesex		3.7216	0.234	15.919	0.000
3.262	4.181				

Niagara		3.8342	0.234	16.399	0.000
3.375	4.294				
Northumberland		4.0509	0.234	17.328	0.000
3.592	4.510				
Ottawa		1.3858	0.234	5.926	0.000
0.926	1.845				
Oxford		2.0384	0.234	8.719	0.000
1.579	2.498				
Peel		2.4382	0.234	10.429	0.000
1.979	2.898				
Perth		0.4472	0.234	1.913	0.056
-0.012	0.906				
Peterborough		0.4992	0.248	2.013	0.045
0.012	0.986				
Prescott and Russell		1.2198	0.234	5.217	0.000
0.760	1.679				
Prince Edward		2.5366	0.234	10.850	0.000
2.077	2.996				
Renfrew		1.5219	0.234	6.508	0.000
1.062	1.981				
Simcoe		3.2236	0.234	13.791	0.000
2.764	3.683				
Stormont, Dundas & Glengarry		2.6483	0.234	11.324	0.000
2.189	3.108				
Thunder Bay		-0.9950	0.248	-4.012	0.000
-1.482	-0.508				
Waterloo		2.5053	0.234	10.717	0.000
2.046	2.965				
Wellington		1.4837	0.234	6.347	0.000
1.024	1.943				
York		1.9513	0.234	8.342	0.000
1.492	2.411				

Omnibus:	40.015	Durbin-Watson:	1.007
Prob(Omnibus):	0.000	Jarque-Bera (JB):	58.845
Skew:	-0.551	Prob(JB):	1.67e-13
Kurtosis:	4.177	Cond. No.	235.

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

3 Specification 3

With County FE + Linear time trend + IV (log_wa_price)

```
[99]: county_dummies = pd.get_dummies(apple_panel['county'], drop_first=True).
      ↪astype(int)

y = apple_panel['log_prod']

exog = sm.add_constant(pd.concat([apple_panel[['log_price', 'trend']],
      ↪county_dummies.reset_index(drop=True)], axis=1))
instr = sm.add_constant(pd.concat([apple_panel[['log_wa_price', 'trend']],
      ↪county_dummies.reset_index(drop=True)], axis=1))

model3 = IV2SLS(y, exog, instr).fit()

print(model3.summary())
```

IV2SLS Regression Results

```
=====
Dep. Variable:          log_prod      R-squared:          0.941
Model:                  IV2SLS        Adj. R-squared:      0.937
Method:                 Two Stage      F-statistic:         201.2
                        Least Squares   Prob (F-statistic):   8.42e-280
Date:                   Tue, 07 Apr 2026
Time:                   20:00:30
No. Observations:      543
Df Residuals:           503
Df Model:               39
=====
```

```
=====
                                coef      std err          t      P>|t|
-----
[0.025      0.975]
-----
const                8.1718      0.516      15.843      0.000
7.158      9.185
log_price            -0.7856      0.144      -5.460      0.000
-1.068     -0.503
trend                0.0453      0.005       8.485      0.000
0.035      0.056
Brant                2.4214      0.212      11.412      0.000
2.005      2.838
Bruce               1.5503      0.212       7.304      0.000
1.133      1.967
Chatham-Kent         2.7306      0.212      12.864      0.000
2.314      3.148
Dufferin            -0.1183      0.212      -0.557      0.578
-0.536      0.299
Durham              4.3968      0.212      20.722      0.000
3.980      4.814
=====
```

Elgin		4.2867	0.212	20.194	0.000
3.870	4.704				
Essex		4.2146	0.212	19.850	0.000
3.797	4.632				
Frontenac		-0.3190	0.214	-1.489	0.137
-0.740	0.102				
Grey		5.2398	0.212	24.658	0.000
4.822	5.657				
Haldimand-Norfolk		4.8216	0.212	22.714	0.000
4.405	5.239				
Halton		2.0558	0.212	9.686	0.000
1.639	2.473				
Hamilton		2.9121	0.212	13.725	0.000
2.495	3.329				
Hastings		1.4269	0.212	6.724	0.000
1.010	1.844				
Huron		1.2634	0.212	5.950	0.000
0.846	1.681				
Kawartha Lakes		-0.2403	0.225	-1.070	0.285
-0.681	0.201				
Lambton		3.4022	0.212	16.019	0.000
2.985	3.819				
Lanark		0.1459	0.212	0.688	0.492
-0.271	0.563				
Leeds & Grenville		1.6715	0.212	7.879	0.000
1.255	2.088				
Lennox & Addington		1.1910	0.214	5.560	0.000
0.770	1.612				
Manitoulin		-0.0592	0.212	-0.279	0.780
-0.476	0.358				
Middlesex		3.5321	0.212	16.642	0.000
3.115	3.949				
Niagara		3.6418	0.212	17.163	0.000
3.225	4.059				
Northumberland		3.8634	0.212	18.198	0.000
3.446	4.280				
Ottawa		1.1913	0.212	5.615	0.000
0.774	1.608				
Oxford		1.8496	0.212	8.713	0.000
1.433	2.267				
Peel		2.2469	0.212	10.588	0.000
1.830	2.664				
Perth		0.2629	0.212	1.237	0.216
-0.154	0.680				
Peterborough		0.3725	0.225	1.659	0.098
-0.069	0.814				
Prescott and Russell		1.0273	0.212	4.841	0.000
0.610	1.444				

Prince Edward		2.3476	0.212	11.060	0.000
1.931	2.765				
Renfrew		1.3266	0.212	6.253	0.000
0.910	1.743				
Simcoe		3.0406	0.212	14.311	0.000
2.623	3.458				
Stormont, Dundas & Glengarry		2.4524	0.212	11.560	0.000
2.036	2.869				
Thunder Bay		-1.1202	0.225	-4.988	0.000
-1.561	-0.679				
Waterloo		2.3191	0.212	10.921	0.000
1.902	2.736				
Wellington		1.2959	0.212	6.104	0.000
0.879	1.713				
York		1.7528	0.212	8.263	0.000
1.336	2.170				

Omnibus:	25.590	Durbin-Watson:	1.224
Prob(Omnibus):	0.000	Jarque-Bera (JB):	35.283
Skew:	-0.406	Prob(JB):	2.18e-08
Kurtosis:	3.949	Cond. No.	620.

3.1 First stage

```
[100]: stage1 = smf.ols('log_price ~ log_wa_price + C(county) + trend',
    ↪data=apple_panel).fit()
print(f"First Stage R-squared: {stage1.rsquared:.6f}")
print(f"F-statistic: {stage1.fvalue:.4f}")
print(f"Kleibergen-Paap F-stat (Stock & Yogo 10% critical): 16.38")
print(f"Instrument is {'STRONG' if stage1.fvalue > 16.38 else 'WEAK'}")
```

```
First Stage R-squared: 0.484331
F-statistic: 12.1136
Kleibergen-Paap F-stat (Stock & Yogo 10% critical): 16.38
Instrument is WEAK
```